

CH iDesign & Renovation — SEO Recommendations

Search Engine Optimisation Assessment & Action Plan

Project	CH iDesign & Renovation — Website Revamp
Document	SEO Assessment & Action Plan
Phase context	Phase 1 complete (frontend, dummy data) → Phase 2 (Firebase, deploy)
Status	Recommendations — not yet implemented
Date	28 May 2026
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Client	Interior Design / Renovation Studio — Batu Kawan, Penang

1. Executive Summary

The Phase 1 site has solid fundamentals already in place: descriptive image `alt` text, lazy-loaded gallery images, a single semantic `<h1>`, a meaningful `<title>`, a base meta description, and font `preconnect` hints. These are the basics many sites get wrong, and they are correct here.

The gaps are **structural** rather than cosmetic. The site is a client-rendered single-page application, so the HTML delivered to crawlers is an empty container; it carries no structured data; it has no social-sharing metadata; and it lacks the `robots.txt` / `sitemap.xml` files search engines expect. For a **local business** competing in a specific geography (Batu Kawan / Penang), these gaps directly limit visibility in exactly the searches that matter most.

This document grades each issue by **impact** and **effort**, then gives a prioritised action plan. The single highest-value, lowest-cost win is adding `LocalBusiness` structured data; the single highest-impact structural change is build-time prerendering so crawlers see real content.

2. What Is Already Correct

Worth stating plainly so it is not undone in future work:

Item	Status	Location
Descriptive <code><title></code>	Good	<code>index.html</code>
Base meta description	Good	<code>index.html</code>
<code>lang="en"</code> on <code><html></code>	Good	<code>index.html</code>
Responsive viewport meta	Good	<code>index.html</code>
Single <code><h1></code> per page	Good	<code>src/components/Hero.jsx</code>
Image alt text	Good	<code>ProjectCard.jsx</code> , <code>ProjectLightbox.jsx</code> , <code>Hero.jsx</code>
Lazy-loaded gallery images	Good	<code>ProjectCard.jsx</code> , <code>ProjectLightbox.jsx</code>
Hero (LCP) image loads eagerly	Good	<code>Hero.jsx</code>
Font <code>preconnect</code> + <code>display=swap</code>	Good	<code>index.html</code>

3. Findings & Recommendations (Prioritised)

Priority key: **P0** = do first (high impact, low effort), **P1** = important, **P2** = polish.

#	Recommendation	Impact	Effort	Priority
1	Build-time prerendering of the public page	High	Medium	P1
2	<code>LocalBusiness</code> structured data (JSON-LD)	High	Low	P0
3	Open Graph + Twitter Card metadata	High	Low	P0
4	<code>robots.txt</code> + <code>sitemap.xml</code>	Medium	Low	P0
5	<code>noindex</code> admin routes	Medium	Low	P0
6	Fix favicon markup + add web manifest	Low	Low	P1
7	Canonical URL	Medium	Low	P1
8	Self-host & optimise project images	Medium	Medium	P1
9	Remove <code>public/.DS_Store</code>	Low	Low	P2
10	LCP hints (<code>fetchpriority</code> , dimensions)	Low	Low	P2

4. Detailed Recommendations

4.1 (P1) Build-time prerendering — *the structural fix*

Problem. `index.html` ships only `<div id="root"></div>`; all content is injected by JavaScript at runtime. Google can render JavaScript, but it does so on a slower second pass and less reliably than static HTML. Crucially, the crawlers behind **link previews** (Facebook, WhatsApp, LinkedIn, X) do **not** execute JavaScript at all — so a shared link currently exposes none of the page's real content.

Fix. Because the public site is effectively a **single page** (/), full server-side rendering is unnecessary. Prerender at build time so `dist/index.html` contains the fully rendered markup:

- `vite-react-ssg` or `react-snap` — minimal change, keeps the current stack; or
- migrate the public page to **Astro** — more capable, larger lift.

This change is what makes every metadata improvement below actually *visible* to crawlers and link-preview bots. Recommended for Phase 2, alongside deployment.

4.2 (Po) `LocalBusiness` structured data — *highest ROI*

Problem. The site emits no structured data, so search engines must infer the business type, location, and contact details from prose.

Fix. Add a JSON-LD `<script type="application/ld+json">` describing the studio as an `InteriorDesign` / `LocalBusiness`. Every field already exists in `src/data/settings.js`:

```
{
  "@context": "https://schema.org",
  "@type": "InteriorDesign",
  "name": "CH iDesign & Renovation",
  "description": "Penang-based interior design and renovation studio.",
  "telephone": "+60 4-123 4567",
  "email": "info@chidr.com.my",
  "address": {
    "@type": "PostalAddress",
    "streetAddress": "Batu Kawan",
    "addressLocality": "Bandar Cassia",
    "postalCode": "14110",
    "addressRegion": "Penang",
    "addressCountry": "MY"
  },
  "openingHours": "Mo-Sa 09:00-18:00",
  "url": "https://<your-domain>",
  "sameAs": ["<facebook>", "<instagram>"]
}
```

This makes the studio eligible for the local pack and rich results — the single most valuable SEO feature for a geographically-focused business.

4.3 (Po) Open Graph + Twitter Card metadata

Problem. There are no `og:*` or `twitter:*` tags, so links shared on social and messaging apps render as bare URLs with no title, description, or image.

Fix. Add to the document head:

```
<meta property="og:type" content="website" />
<meta property="og:title" content="CH iDesign & Renovation – Interior Design Studio, Penang" />
<meta property="og:description" content="Spaces that inspire the way you live. Interior design" />
<meta property="og:image" content="https://<your-domain>/CHIR.png" />
<meta property="og:url" content="https://<your-domain>/" />
<meta name="twitter:card" content="summary_large_image" />
```

React 19 hoists `<title>` and `<meta>` rendered inside components, so these can be managed per-route without a helmet library once the site grows.

4.4 (Po) robots.txt + sitemap.xml

Problem. Neither file exists. Search engines have no sitemap to crawl and no crawl directives.

Fix. Add both to `public/` (Vite copies `public/` to the build root):

```
# public/robots.txt
User-agent: *
Allow: /
Disallow: /admin
Sitemap: https://<your-domain>/sitemap.xml
```

```
<!-- public/sitemap.xml -->
<urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9">
  <url><loc>https://<your-domain>/</loc></url>
</urlset>
```

4.5 (Po) noindex the admin area

Problem. The `/admin` routes are crawlable. The demo gate is cosmetic, so nothing stops the admin UI from being indexed.

Fix. Two layers: `Disallow: /admin` in `robots.txt` (above), and a `<meta name="robots" content="noindex, nofollow">` rendered on admin pages (e.g. in `AdminLayout / Login`).

4.6 (P1) Favicon markup + web manifest

Problem. `index.html` declares `type="image/svg+xml"` while pointing at `favicon.ico` — a mismatch. The repo already contains `favicon.svg`, a `favicons/` folder, and `logo.svg` that are not fully wired up.

Fix. Reference the SVG favicon correctly, add an `apple-touch-icon`, and add a `site.webmanifest` for installability and richer mobile/search presentation.

4.7 (P1) Canonical URL

Problem. No `<link rel="canonical">`, which risks duplicate-URL dilution (e.g. with/without trailing slash, `www` vs apex) once deployed.

Fix. Add `<link rel="canonical" href="https://<your-domain>/" />` once the production domain is fixed.

4.8 (P1) Self-host & optimise project images

Problem. Project covers are remote Unsplash URLs. They depend on a third party, are not format-optimised, and contribute to Largest Contentful Paint — a Core Web Vitals ranking signal.

Fix. In Phase 2, store optimised WebP/AVIF via the planned Cloudinary pipeline, serve responsive sizes, and keep dimensions explicit to avoid layout shift (CLS).

4.9 (P2) Remove `public/.DS_Store`

Problem. `public/.DS_Store` is a macOS artefact that Vite will copy into the deployed build.

Fix. Delete it and add `.DS_Store` to `.gitignore`.

4.10 (P2) LCP hints

Problem. The hero image is the Largest Contentful Paint element but carries no priority hint or explicit dimensions.

Fix. Add `fetchpriority="high"` and explicit `width / height` (or an aspect ratio) to the hero `<img>` in `Hero.jsx`.

5. Suggested Sequencing

Now (Phase 1 polish — no rendering changes, ~1 hour):

- `LocalBusiness` JSON-LD (§4.2)
- Open Graph + Twitter tags (§4.3)
- `robots.txt` + `sitemap.xml` (§4.4)
- `noindex` admin (§4.5)
- Favicon + manifest (§4.6)
- Remove `.DS_Store` (§4.9)
- LCP hints (§4.10)

Phase 2 (with deployment & Firebase):

- Build-time prerendering (§4.1) — depends on a known production setup
- Canonical URL (§4.7) — depends on a fixed domain
- Self-hosted / optimised images via Cloudinary (§4.8)

6. Measuring Success

After deployment, verify and monitor with:

Tool	Checks
Google Search Console	Indexing, queries, sitemap submission, rich-result eligibility
Rich Results Test	Validates LocalBusiness JSON-LD
PageSpeed Insights / Lighthouse	Core Web Vitals (LCP, CLS, INP), SEO score
Facebook Sharing Debugger / X Card Validator	Open Graph / Twitter preview rendering
Google Business Profile	Local pack presence (complements on-site SEO)

Note: On-site SEO is necessary but not sufficient for a local studio. A verified, complete **Google Business Profile** — with the same name, address, and phone as the JSON-LD above — typically drives more local enquiries than any single on-page change. Keep the two consistent.